

DFIR ANALYSIS REPORT

Full Correlated Investigation — Revised

Field	Value
Case Numbers	Client_Forensics (0001) Windows_Final (12345)
Devices	BGOC111 (Client, Windows 10 Education) BGOC00 (Server, Windows Server 2022 Datacenter)
Domain	bgoc.corp
Timezone	Asia/Baku (AZT, UTC+4)
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Report Date	2026-06-04

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1. Executive Summary

A multi-stage, multi-actor compromise affected at least two bgoc.corp domain-joined hosts — the Windows 10 client BGOC111 and the Windows Server 2022 host BGOC00 — with artefacts spanning from November 2023 through November 2025. Cross-referencing both forensic images produces a coherent end-to-end attack chain: initial access via a social-engineered WhatsApp payload delivered to an end user on the client, tooling deployment, Active Directory reconnaissance, lateral movement to the domain server, credential harvesting, data collection, and culminating in data exfiltration and deliberate anti-forensic evidence destruction on the server.

The shared domain Administrator SID (S-1-5-21-4227448782-3866209265-969814615-500) appears in artefacts on both machines, confirming the same privileged account was used across both hosts. The account groupf is active on both systems simultaneously on 2025-11-04 — running attack tools on the client while accessing staged data on the server — making it the primary hands-on-keyboard operator account for the final phase of the attack.

2. Server Profile & Confirmed Network Topology

Field	Value
Hostname	BGOC00
OS	Windows Server 2022 Datacenter (Build 20348)
Domain	bgoc.corp
IP Address	192.168.80.161
Platform	VMware Virtual Machine

The following registry artefacts confirm the client OS profile (Windows 10 Education, Released 2009, CurrentVersion 6.3):

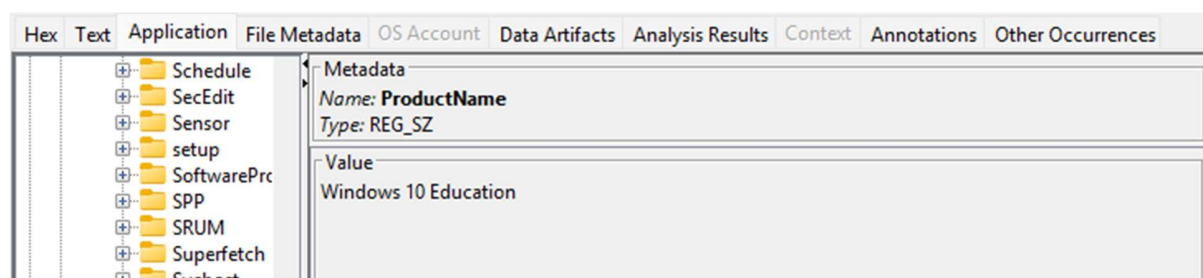


Figure 1a: Autopsy Registry Viewer — HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion: ProductName = Windows 10 Education, confirming client OS edition.

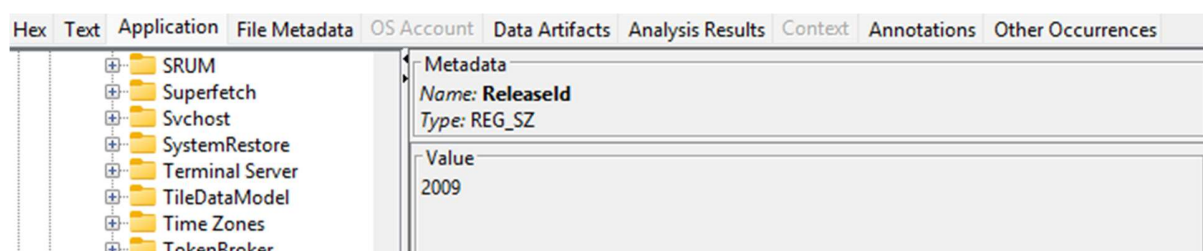
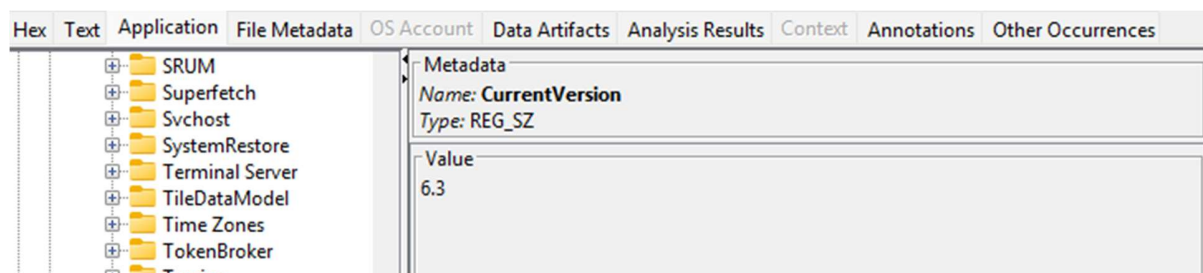


Figure 1b: Autopsy Registry Viewer — *ReleaseId = 2009, identifying the Windows 10 feature update version.***Figure 1c:** Autopsy Registry Viewer — *CurrentVersion = 6.3, the internal NT kernel version string for Windows 10.*

Three distinct hosts are involved in the attack infrastructure. Correctly separating these is critical for correlation:

IP / Hostname	Role	How Confirmed
192.168.80.161	BGOC00 — the imaged server	Server Administrator browsed http://192.168.80.161:9090/ (self-referential)
192.168.80.65	Attacker staging server (not imaged)	Client: Invoke-PowerShellTcp.ps1 downloaded from this host on port 8888
\\DC (hostname)	Domain Controller (not imaged)	Client shell bags: \\dc\\c\$, \\dc\\SYSVOL, \\dc\\TOOLS

Confirmed accounts on BGOC00: Administrator, raul, cbell, groupf, ohughes, etaylor, obell, ruby.rozamond, groupx, groupa, lcooper.

3. Initial Access & Execution

3.1 Primary Entry Point — WhatsApp Social Engineering (Client, 2024-10-25)

The initial compromise of the environment began with user john on BGOC111 receiving malicious files through WhatsApp Web on 2024-10-25. The delivery sequence, taken directly from Chrome download history, was as follows:

Time (AZT)	File	Technique
10:32	capcha verification.bat	Fake CAPTCHA / ClickFix lure
10:38	salam.txt	Decoy benign file to reduce suspicion
10:57	capcha verification (1).bat	Re-delivery
11:06	thm_voucher.zip → thm_voucher.txt.exe	Double-extension executable, Internet Zone marked

History	1	C:\Users\john\Downloads\capcha verification.bat	blob:https://web.whatsapp.com/64abdcfa-817e-4a89-...
History	1	C:\Users\john\Downloads\salam.txt	blob:https://web.whatsapp.com/bae32e4e-777f-4ec9-...
History	1	C:\Users\john\Downloads\capcha verification (1).bat	blob:https://web.whatsapp.com/44c82628-393e-4ee8-...

Figure 3a: Autopsy Web Downloads — Chrome download history filtered by user john, showing captcha verification.bat, salam.txt, and captcha verification (1).bat downloaded from blob:https://web.whatsapp.com/ URLs, confirming WhatsApp Web as the delivery vector.

History	1	C:\Users\john\Downloads\thm_voucher.zip	blob:https://web.whatsapp.com/74cba4dc-e49f-4d50-...
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Figure 3b: Autopsy Web Downloads — Fourth download entry showing thm_voucher.zip, also sourced from a WhatsApp Web blob URL, completing the four-file delivery chain.

History	1	C:\Users\msilva\Downloads\Safari_3.0.1_beta.exe	https://downloads.sourceforge.net/project/virtualhac...
History	1	C:\Users\msilva\Downloads\Safari_3.0.1_beta.exe	https://master.dl.sourceforge.net/project/virtualhack...
thm_voucher.txt.exe:Zone.Identifier		/Users/john/Downloads/thm_voucher/thm_voucher.t...	
History	1		https://external-preview.redd.it/uhq5zTcMPM3tOW_f...
History	1		https://external-preview.redd.it/uhq5zTcMPM3tOW_f...

Figure 4: Autopsy Web Downloads — thm_voucher.txt.exe visible alongside its Zone.Identifier alternate data stream entry, confirming Internet Zone marking and exposing the double-extension masquerade designed to appear as a text file in Windows Explorer.

The .bat files named "captcha verification" are consistent with the ClickFix technique, which instructs the victim to run an attacker-controlled script under the pretence of a CAPTCHA verification step. The thm_voucher.txt.exe abuses Windows Explorer's default behaviour of hiding known file extensions, making the file appear to be a .txt document to an untrained user.

3.2 Client Precursor Activity (2023-11-20)

The earliest suspicious execution cluster on BGOC111 at approximately 23:10 on 2023-11-20 shows WHOAMI.EXE, CERTUTIL.EXE (four executions in rapid succession), NET.EXE, NET1.EXE, SHTASKS.EXE, and the non-standard DACLSERVICE.EXE from C:\Program Files\DACL Service\. This cluster at a late-night timestamp predates the WhatsApp delivery by approximately eleven months and suggests an earlier, separate access method established the initial foothold.

CERTUTIL.EXE	CERTUTIL.EXE	2023-11-20 23:10:33 AZT	Prefetch File
CERTUTIL.EXE	CERTUTIL.EXE	2023-11-20 23:10:21 AZT	Prefetch File
CERTUTIL.EXE	CERTUTIL.EXE	2023-11-20 23:10:25 AZT	Prefetch File
CERTUTIL.EXE	CERTUTIL.EXE	2023-11-20 23:10:29 AZT	Prefetch File
CERTUTIL.EXE	CERTUTIL.EXE	2023-11-20 23:10:33 AZT	Prefetch File
CERTUTIL.EXE	CERTUTIL.EXE	2023-11-20 23:10:21 AZT	Prefetch File
CERTUTIL.EXE	CERTUTIL.EXE	2023-11-20 23:10:25 AZT	Prefetch File
CERTUTIL.EXE	CERTUTIL.EXE	2023-11-20 23:10:29 AZT	Prefetch File

Figure 5a: Autopsy Run Programs / Prefetch — CERTUTIL.EXE executed four times in rapid succession on 2023-11-20 between 23:10:21 and 23:10:33 AZT, a pattern consistent with staged payload retrieval or certificate manipulation.

whoami.exe	domainadmin
	enterpriseadmin
	localservice
	networkservice

Figure 5b: Autopsy Prefetch keyword view — WHOAMI.EXE highlighted, confirming execution at the same 2023-11-20 23:10 cluster alongside domainadmin, enterpriseadmin, localservice, and networkservice strings, indicating identity and privilege enumeration.

SCHTASKS.EXE	SCHTASKS.EXE	2023-11-20 23:10:41 AZT	Prefetch File
SCHTASKS.EXE	SCHTASKS.EXE	2023-11-20 23:10:41 AZT	Prefetch File

Figure 5c: Autopsy Run Programs — SCHTASKS.EXE executed twice at 2023-11-20 23:10:41 AZT, indicating scheduled task creation or enumeration as part of the same early-access activity cluster.

3.3 Client Machine Prepared as Attack Platform (2024-03-07)

A large batch of tools was installed on a single morning on 2024-03-07, suggesting the machine was either reimaged or formally configured as an attack platform on that date. All of the following were installed within the same window: Nmap 7.93, Wireshark 4.2.3, Npcap 1.73, Burp Suite Community 2022.12.4, AutoHotkey, KeePass Password Safe 2.56, WinSCP 6.1.1, RealVNC Viewer, OpenVPN 2.6.9, 7-Zip, HxD Hex Editor, DB Browser for SQLite, AWS CLI, Microsoft Azure CLI 2.58.0, and multiple .NET and Java runtimes. Chocolatey was used as the package manager.

SOFTWARE	0	AutoHotkey 1.1.37.01 v.1.1.37.01	2024-03-07 09:11:01 AZT	Client-Image.E01
SOFTWARE	0	AutoHotkey 1.1.37.01 v.1.1.37.01	2024-03-07 09:11:01 AZT	Client-Image.E01
SOFTWARE	0	Npcap v.1.73	2024-03-07 09:11:27 AZT	Client-Image.E01
SOFTWARE	0	Npcap v.1.73	2024-03-07 09:11:27 AZT	Client-Image.E01
SOFTWARE	0	Wireshark 4.2.3 x64 v.4.2.3	2024-03-07 09:12:06 AZT	Client-Image.E01
SOFTWARE	0	Wireshark 4.2.3 x64 v.4.2.3	2024-03-07 09:12:06 AZT	Client-Image.E01
SOFTWARE	0	Burp Suite Community Edition 2022.12.4 v.2022.12.4	2024-03-07 09:22:16 AZT	Client-Image.E01
SOFTWARE	0	Burp Suite Community Edition 2022.12.4 v.2022.12.4	2024-03-07 09:22:16 AZT	Client-Image.E01
SOFTWARE	0	HxD Hex Editor 2.5 v.2.5	2024-03-07 09:26:28 AZT	Client-Image.E01
SOFTWARE	0	HxD Hex Editor 2.5 v.2.5	2024-03-07 09:26:28 AZT	Client-Image.E01
SOFTWARE	0	Nmap 7.93 v.7.93	2024-03-07 09:31:50 AZT	Client-Image.E01
SOFTWARE	0	Nmap 7.93 v.7.93	2024-03-07 09:31:50 AZT	Client-Image.E01
SOFTWARE	0	RealVNC Viewer 7.7.0 v.7.7.0.51227	2024-03-07 09:49:37 AZT	Client-Image.E01
SOFTWARE	0	RealVNC Viewer 7.7.0 v.7.7.0.51227	2024-03-07 09:49:37 AZT	Client-Image.E01
SOFTWARE	0	7-Zip 23.01 (x64) v.23.01	2024-03-07 09:52:37 AZT	Client-Image.E01
SOFTWARE	0	7-Zip 23.01 (x64) v.23.01	2024-03-07 09:52:37 AZT	Client-Image.E01
SOFTWARE	0	OpenVPN 2.6.9-1001 amd64 v.2.6.901	2024-03-07 09:53:47 AZT	Client-Image.E01
SOFTWARE	0	OpenVPN 2.6.9-1001 amd64 v.2.6.901	2024-03-07 09:53:47 AZT	Client-Image.E01
SOFTWARE	0	DB Browser for SQLite v.3.12.2	2024-03-07 09:55:12 AZT	Client-Image.E01

Figure 6: Autopsy Installed Programs — Registry SOFTWARE hive records showing the mass tooling deployment on 2024-03-07. AutoHotkey, Npcap, Wireshark 4.2.3, Burp Suite Community 2022.12.4, HxD Hex Editor, Nmap 7.93, RealVNC Viewer 7.7.0, 7-Zip, OpenVPN 2.6.9, and DB Browser for SQLite all installed within a two-hour window, confirming deliberate attack platform configuration.

3.4 Server — Earliest Confirmed Activity & WeakDomain.ps1

The server (BGOC00) entered the domain at some point prior to 2024-11-29. A PowerShell script named WeakDomain.ps1 is present in C:\Users\Administrator\Downloads\. The Recent Documents artefact for this file was last updated 2024-12-02 10:05:11 AZT. It is important to note that Recent Document timestamps reflect when the LNK or JumpList record was last written — this does not necessarily equal the file's creation or original download date, and the actual arrival of this script on the server may predate December 2024.

3.5 Server Compromise May Predate the Main Attack by Months

Python 3.13.3 was installed on BGOC00 on 2025-05-30, and a file called generate.log in C:\Users\Administrator\Downloads\ was last accessed on 2025-06-09. This places scripted activity on the server approximately five months before the main November 2025 attack window, meaningfully pushing back the server compromise timeline beyond what the 2025-11-04 artefacts alone would suggest.

Keyword search4 Results

TableThumbnailSummary

Save Table as CSV

Name	Keyword Preview	Location	Modified Time	Change T
 DACLSERVICE.EXE-3323C0F5.pf-slack	«daclservice.exe»-3323c0f5.pf-slack	/img_Client-Image.E01/vol_vol6/Windows/Prefetch/D...	2023-11-20 23:10:32 AZT	2024-03-0
 DACLSERVICE.EXE-3323C0F5.pf	«daclservice.exe»-3323c0f5.pf	/img_Client-Image.E01/vol_vol6/Windows/Prefetch/D...	2023-11-20 23:10:32 AZT	2024-03-0
 daclservice.exe-slack	«daclservice.exe»-slack	/img_Client-Image.E01/vol_vol6/Program Files/DACL ...	2023-11-20 23:10:21 AZT	2023-11-2
 daclservice.exe	«daclservice.exe»	/img_Client-Image.E01/vol_vol6/Program Files/DACL ...	2023-11-20 23:10:21 AZT	2023-11-2

Figure 9: Autopsy Keyword Search — DACLSERVICE.EXE artefact results showing the prefetch file (DACLSERVICE.EXE-3323C0F5.pf) and binary (daclservice.exe) in C:\Program Files\DACL Service\, modified 2023-11-20 23:10:21 AZT, establishing the earliest confirmed backdoor timestamp on the client.

4. Persistence & Command-and-Control

4.1 Autorun Persistence Binary (Client)

C:\Program Files\Autorun Program\program.exe executed twice on the client on 2025-11-04 at 09:48:57 and 11:42:54 AZT. This is not a standard Windows or known third-party application. Its location in Program Files implies installation with administrative privileges, likely via a scheduled task or service entry. This file should be retrieved, hashed, and submitted for malware analysis as the primary persistence mechanism on the client.

4.2 DACL Service Backdoor (Client)

C:\Program Files\DACL Service\daclservice.exe executed once on 2023-11-20 at 23:10:22 — the earliest confirmed backdoor artefact in the environment.

4.3 Atera RMM Agent (Client)

Autopsy flagged C:\ProgramData_VM\log.txt as an Atera artefact. Atera is a commercial Remote Monitoring and Management platform providing persistent, encrypted remote access that blends with legitimate IT management traffic.

Listing Keyword search 1 - WHOAMI.EXE × | Keyword search 2 - WHOAMI.EXE CER... × | Keyword search 3 - CERTUTIL.EXE × | Keyword search 4 - ... ×

Remote Monitoring Management Programs 31 Results

Table Thumbnail Summary

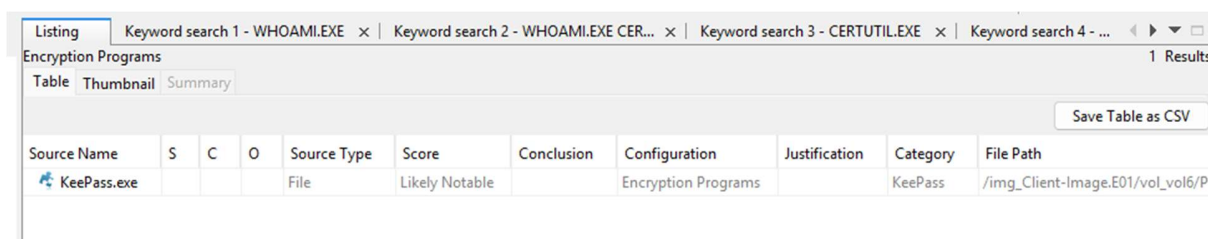
Save Table as CSV

Source Name	S	C	O	Source Type	Score	Conclusion	Configuration	Justification	Category
quickassist.exe				File	Likely Notable		Remote Monitoring Management Programs		Microsoft Quick Assist
mstsc.exe				File	Likely Notable		Remote Monitoring Management Programs		mstsc
mstsc.exe				File	Likely Notable		Remote Monitoring Management Programs		mstsc
dd.exe				File	Likely Notable		Remote Monitoring Management Programs		Distant Desktop
mstsc.exe				File	Likely Notable		Remote Monitoring Management Programs		mstsc
mstsc.exe				File	Likely Notable		Remote Monitoring Management Programs		mstsc
mstsc.exe				File	Likely Notable		Remote Monitoring Management Programs		mstsc
quickassist.exe				File	Likely Notable		Remote Monitoring Management Programs		Microsoft Quick Assist
mstsc.exe				File	Likely Notable		Remote Monitoring Management Programs		mstsc
quickassist.exe				File	Likely Notable		Remote Monitoring Management Programs		Microsoft Quick Assist
mstsc.exe				File	Likely Notable		Remote Monitoring Management Programs		mstsc
winscp.exe				File	Likely Notable		Remote Monitoring Management Programs		WinSCP
quickassist.exe				File	Likely Notable		Remote Monitoring Management Programs		Microsoft Quick Assist
connect.exe				File	Likely Notable		Remote Monitoring Management Programs		Zoho Assist
dd.exe				File	Likely Notable		Remote Monitoring Management Programs		Distant Desktop
user.xml				File	Likely Notable		Remote Monitoring Management Programs		ScreenConnect
user.config				File	Likely Notable		Remote Monitoring Management Programs		mRemoteNG
PSEXEC.exe				File	Likely Notable		Remote Monitoring Management Programs		PSEXEC
config.json				File	Likely Notable		Remote Monitoring Management Programs		Bluetrait
config.json				File	Likely Notable		Remote Monitoring Management Programs		Bluetrait
user.config				File	Likely Notable		Remote Monitoring Management Programs		mRemoteNG
log.txt				File	Likely Notable		Remote Monitoring Management Programs		Atera
dd.exe				File	Likely Notable		Remote Monitoring Management Programs		Distant Desktop
config.json				File	Likely Notable		Remote Monitoring Management Programs		Bluetrait
quickassist.exe				File	Likely Notable		Remote Monitoring Management Programs		Microsoft Quick Assist
vncviewer.exe				File	Likely Notable		Remote Monitoring Management Programs		VNC
mstsc.exe				File	Likely Notable		Remote Monitoring Management Programs		mstsc
quickassist.exe				File	Likely Notable		Remote Monitoring Management Programs		Microsoft Quick Assist
quickassist.exe				File	Likely Notable		Remote Monitoring Management Programs		Microsoft Quick Assist
quickassist.exe				File	Likely Notable		Remote Monitoring Management Programs		Microsoft Quick Assist
WinSCP.exe				File	Likely Notable		Remote Monitoring Management Programs		WinSCP

Figure 12: Autopsy Interesting Items — Remote Monitoring and Management Programs category showing 31 flagged results including Atera (log.txt), mRemoteNG (user.config), WinSCP, Microsoft Quick Assist (quickassist.exe), Distant Desktop (dd.exe), VNC (vncviewer.exe), PSEXEC, and Bluetrait (config.json). This breadth of RMM tooling confirms multiple remote-access channels were maintained on the client.

4.4 Nishang Reverse Shell — Staging Server Confirmed (Client, 2025-04-30)

The account administrator.BGOC on the client downloaded Invoke-PowerShellTcp.ps1 from <http://192.168.80.65:8888/Invoke-PowerShellTcp.ps1> twice on 2025-04-30 at 14:49 and 15:03 AZT. Invoke-PowerShellTcp is part of the Nishang PowerShell attack framework and establishes a raw TCP reverse shell to an attacker-specified listener. This is a distinct mechanism from WinRM/PowerShell Remoting — it operates over arbitrary TCP, not ports 5985/5986.



The screenshot shows the Autopsy interface with the 'Encryption Programs' category selected. The table lists the following results:

Source Name	S	C	O	Source Type	Score	Conclusion	Configuration	Justification	Category	File Path
KeePass.exe				File	Likely Notable		Encryption Programs		KeePass	/img_Client-Image.E01/vol_vol6/P

Figure 13: Autopsy Interesting Items — Encryption Programs category flagging KeePass.exe at its installation path on the client image. KeePass is a password manager; in an attack context, the attacker may target the KeePass database to harvest stored credentials.

On the server, wsmprovhos.exe (the WinRM PowerShell Remoting host process) executed at 2025-11-04 13:00:48 AZT, indicating an inbound PowerShell Remoting session was received by BGOC00 at that time. Whether the attacker chained these mechanisms — using the Nishang shell on the client to then initiate a WinRM session to the server — is a reasonable hypothesis but is not directly proven by the artefacts in these two images alone.

4.5 Port 9090 on BGOC00 — Unidentified Web Service

The Administrator's browser history shows <http://192.168.80.161:9090/> browsed at 11:27:20 AZT on 2025-11-04 — the server accessing its own IP on a non-standard port. Something is serving HTTP on port 9090. This warrants investigation to identify the process. Port 9090 hosts a wide range of legitimate and attacker-related software, so no assumption about its nature should be made without reviewing the installed services and process list from a memory dump or live system.

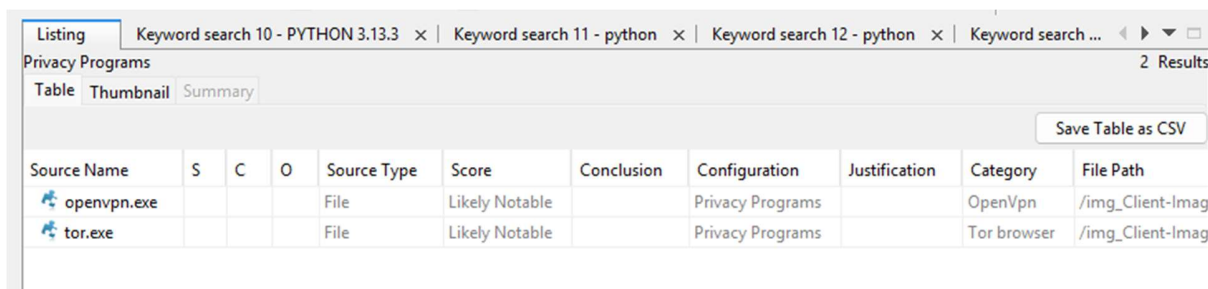
<http://lockyourdevice.com/> was browsed eight seconds later. This is commercially a device management service and should not be labelled as attacker infrastructure based solely on timing without further corroborating evidence.

4.6 C3 Covert C2 Framework (Client)

The `C:\Tools\C3\` directory contains implementation guides for building C2 channels through Office 365 and Azure APIs. C3 routes command-and-control traffic through legitimate Microsoft cloud service requests, making it effectively invisible to perimeter inspection.

4.7 Tor Browser & OpenVPN (Client)

Tor Browser and OpenVPN are both installed, providing covert egress channels. Tor Browser is installed via Chocolatey and OpenVPN is present in Program Files, both providing multiple layers of anonymisation and encrypted tunnelling capability.



The screenshot shows the Autopsy interface with the 'Privacy Programs' category selected. The table lists the following results:

Source Name	S	C	O	Source Type	Score	Conclusion	Configuration	Justification	Category	File Path
openvpn.exe				File	Likely Notable		Privacy Programs		OpenVpn	/img_Client-Imag
tor.exe				File	Likely Notable		Privacy Programs		Tor browser	/img_Client-Imag

Figure 17: Autopsy Interesting Items — Privacy Programs category flagging `openvpn.exe` (OpenVPN) and `tor.exe` (Tor Browser) on the client image. Both are rated Likely Notable and confirm covert egress capability beyond standard browser-based exfiltration.

4.8 mRemoteNG — Flagged by Autopsy, Requires Confirmation

Autopsy's Interesting Items module flagged user.config files in .NET isolated storage paths on both machines. The parent application names (`lbfoadmin.exe`, `ServerManager.exe`, `PowerShell_ISE.exe`) are standard Windows binaries, not mRemoteNG itself. The Autopsy rule appears to match on the isolated storage pattern rather than confirmed mRemoteNG content. These files must be opened and their XML contents inspected to confirm whether they contain mRemoteNG connection data before treating these as confirmed findings.

5. Privilege Escalation & Lateral Movement

5.1 Credential Dumping — Mimikatz (Client, 2025-11-04)

Mimikatz executed on the client at 12:01:00 and 13:03:00 AZT from two separate locations simultaneously:

- `C:\Tools\Mimikatz\x64\mimikatz.exe`
- `C:\Users\groupf\Desktop\mimikatz.exe`

The 12:01:00 execution coincides with `whoami.exe`, `net1.exe`, and `dsregcmd.exe` also running — a classic post-exploitation sequence of identity verification, user/group enumeration, and domain-join/Azure AD state checks.

mimikatz.exe	OpenThreadToken (0x%08x)«whoami.exe«domainad...	/img_Client-Image.E01/vol_vol6/Users/groupf/Desktop... 2025-1
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Figure 19: Autopsy Run Programs / Interesting Items — `mimikatz.exe` entry showing the file path `C:\Users\groupf\Desktop\mimikatz.exe` with an `OpenThreadToken` Windows API call context string, confirming execution under a domain administrator token impersonation context on 2025-11-04.

5.2 Kerberos Ticket Artefact — Domain-Level Authentication Compromise

`C:\Tools\Mimikatz\x64\[0;4a5971]-2-1-40e10000-john@krbtgt-ACC.LAB.kirbi` is a Kerberos TGT export for the john account on the ACC.LAB domain targeted at the `krbtgt` service. The `.kirbi` format is produced by Mimikatz's `kerberos::list /export` or `sekurlsa::tickets /export` commands. A ticket targeting `krbtgt` is the signature of either a Golden Ticket attack or pass-the-ticket reuse of a harvested TGT. ACC.LAB is a separate domain from bgoc.corp — both domains should be treated as compromised.

5.3 Active Directory Enumeration — BloodHound & SharpHound (Both Machines)

BloodHound output from a SharpHound collection against the ACC domain was stored on the client at `C:\Users\msilva\Desktop\Acc_Domain_20240405092029_BloodHound\Acc_Domain_20240405092029_users.json`, timestamped 2024-04-05 09:20:29 AZT. This file was reviewed by the nihat account on 2025-04-30.



Figure 22a: Autopsy Interesting Items — BloodHound.exe flagged in the client image at C:\img_Client-Image.E01\vol_vol6\Tools\BloodHound\, last changed 2023-05-24 18:51:26 AZT (install date 2024-03-07). BloodHound is the primary Active Directory attack path visualisation tool, confirming tooling readiness for AD enumeration.

BloodHound.Ink

\\DC\TOOLS\BloodHound

0000-00-00 00:00:00

Client-Image.E01

Figure 22b: Autopsy Recent Documents / Shell Bags — BloodHound.lnk record from msilva's AutomaticDestinations JumpList, showing the target path \\DC\TOOLS\BloodHound. This confirms msilva accessed the BloodHound installation stored on the Domain Controller's TOOLS share, establishing a link between the client user and the DC tool repository.

Data Sources

- Client-Image.E01_1 Host
 - Client-Image.E01
 - vol1 (Unallocated: 0-2047)
 - vol4 (EFI system partition: 2048-411647)
 - vol5 (Microsoft reserved partition: 411648-673792)
 - vol6 (Basic data partition: 673792-146798)
 - \$ORphanFiles (91)
 - \$CarvedFiles (1)
 - \$Extend (9)
 - \$Recycle.Bin (13)
 - \$NtfsLog (35)
 - commando (11)
 - Documents and Settings (2)
 - Missing Scheduled Binary (2)
 - PerfLogs (2)
 - Program Files (68)
 - Program Files (x86) (43)
 - ProgramData (26)
 - Recovery (3)
 - System Volume Information (6)
 - Temp (13)
 - Tools (85)

Listing Keyword search 1 - WHOAMI.EXE x Keyword search 2 - WHOAMI.EXE CER... x Keyword search 3 - CERTUTIL.EXE x Keyword search 4 - WHO... x

/img_Client-Image.E01/vol_vol6/Users/msilva/Desktop

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
[current folder]				2025-04-30 15:24:01 AZT	2025-04-30 15:24:01 AZT	2025-05-06 18:56:27 AZT	2024-02-27 16:47:05 AZT
[parent folder]				2024-04-20 23:35:08 AZT	2025-04-30 15:23:52 AZT	2025-11-04 11:34:16 AZT	2024-02-27 16:47:05 AZT
Acc Domain_20240405092029_BloodHound				2025-04-30 15:24:01 AZT	2025-04-30 15:24:01 AZT	2025-05-06 18:56:27 AZT	2025-04-30 15:24:01 AZT
Acc Domain_20240405092029_BloodHound.zip			0	2024-04-05 10:20:30 AZT	2025-04-30 15:23:52 AZT	2025-04-30 15:24:01 AZT	2024-04-05 10:20:30 AZT
desktop.ini			1	2024-02-27 16:47:15 AZT	2025-04-30 15:23:52 AZT	2025-05-06 18:56:06 AZT	2024-02-27 16:47:15 AZT
NzhjNGU3MWltNmRmOC00ZWMLTk0MTktkNzZl			0	2024-04-05 10:20:30 AZT	2025-04-30 15:23:52 AZT	2024-04-05 10:20:30 AZT	2024-04-05 10:20:30 AZT

Save Table as CSV

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Type	Value	Source(s)	Recent Activity
Associated Artifact	-9223372036854775450		
Source File Path	/img_Client-Image.E01/vol_vol6/Users/msilva/Desktop/Acc Domain_20240405092029_BloodHound		
Artifact ID	-9223372036854775449		

Figure 22c: Autopsy Directory Listing — C:\Users\msilva\Desktop\ showing the Acc Domain_20240405092029_BloodHound folder and its associated .zip archive. The folder modification timestamp of 2025-04-30 15:24:01 AZT indicates nihat reviewed this BloodHound collection output approximately a year after it was created.

A completely separate, fresh SharpHound v2.8.0 collection was initiated directly from BGOC00 on 2025-11-04 at 11:55:37 AZT when the Administrator account downloaded it from the SpecterOps GitHub repository. The zip was opened at 11:57:12 and the resulting output was almost certainly packaged into test.zip for exfiltration.

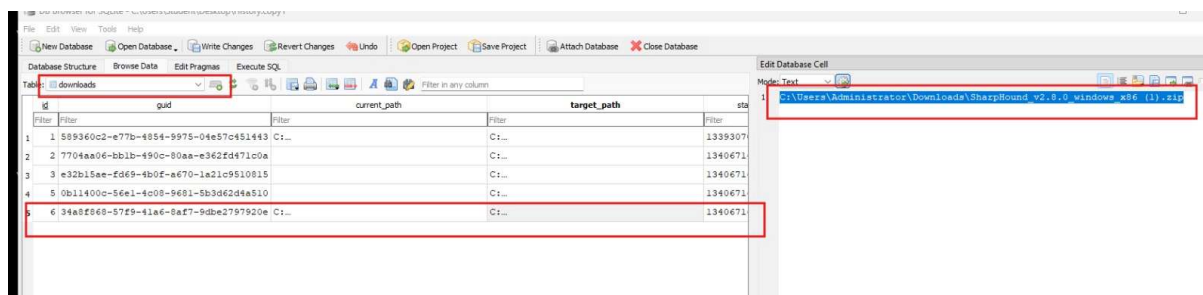


Figure 24: DB Browser for SQLite — Edge browser download database (history.copy) opened on the server image. The highlighted row (id=6) in the downloads table shows target_path: C:\Users\Administrator\Downloads\SharpHound_v2.8.0_windows_x86(1).zip, confirming the Administrator account downloaded SharpHound v2.8.0 directly from GitHub on 2025-11-04.

5.4 Domain Controller Access Confirmed (Client Shell Bags)

Shell bag artefacts from the client confirm direct UNC share access to the DC:

User	Path Accessed	Significance
john	\\dc\c\$	Administrative share — requires Domain Admin or equivalent
john	\\dc\SYSVOL\acc.lab	SYSVOL for ACC.LAB domain
msilva	\\dc\Tools	Attacker toolkit share hosted on DC
msilva	\\dc\Tools\ADConnectDump	Azure AD Connect credential extraction tool

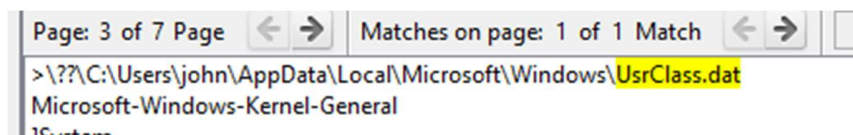


Figure 26a: Autopsy Text View — john's UsrClass.dat shell bag database located at C:\Users\john\AppData\Local\Microsoft\Windows\UsrClass.dat, confirmed as the source file for the UNC path shell bag artefacts. This file records john's navigation history including the \\DC\c\$ and \\DC\SYSVOL\ACC.LAB paths.

Listing Keyword search 10 - PYTHON 3.13.3 Keyword search 11 - python Keyword search 12 - python Keyword search ... 17 Results

Table Thumbnail Summary Save Table as CSV

Name	S	C	O	Login Name	Host	Scope	Realm Name	Creation Time
S-1-5-80-956008885-3418522649-1831038044-18532			0		Client-Im...	Local	NT SERVICE	
S-1-5-18				SYSTEM	Client-Im...	Local	NT AUTHORITY	
S-1-5-80-3028837079-3186095147-955107200-37019			0		Client-Im...	Local	NT SERVICE	
S-1-5-21-1485720640-942268468-2087739954-2104			0	msilva	Client-Im...	Domain		
S-1-5-21-4227448782-3866209265-969814615-1457			0	groupf	Client-Im...	Domain		
S-1-5-21-2106596318-82463151-3463888505-1005			0	zorkete	Client-Im...	Domain		
S-1-5-19				LOCAL SERVICE	Client-Im...	Local	NT AUTHORITY	
S-1-5-21-2106596318-82463151-3463888505-1000			0	user	Client-Im...	Domain		
S-1-5-21-1485720640-942268468-2087739954-1177			0	adiana	Client-Im...	Domain		
S-1-5-21-2106596318-82463151-3463888505-500			0	administrator	Client-Im...	Domain		
S-1-5-21-1485720640-942268468-2087739954-500			0	administrator	Client-Im...	Domain		
S-1-5-21-4227448782-3866209265-969814615-500			0	administrator	Client-Im...	Domain		
S-1-5-21-1485720640-942268468-2087739954-2221			0	john	Client-Im...	Domain		
S-1-5-21-2106596318-82463151-3463888505-1004			0	nihat	Client-Im...	Domain		
S-1-5-21-1485720640-942268468-2087739954-1189			0	kurt	Client-Im...	Domain		
S-1-5-20				NETWORK SERVICE	Client-Im...	Local	NT AUTHORITY	
S-1-5-21-3933942852-973373972-2766786355-1032			0		Client-Im...	Domain		

Figure 26b: Autopsy OS Accounts — Domain account SID listing from the client image showing all domain-authenticated accounts: msilva, groupf, zorkete, user, adiana, administrator (multiple SIDs from different domains), john, nihat, and kurt. The presence of multiple administrator SIDs from different domains (S-1-5-21-1485720640 and S-1-5-21-4227448782) confirms cross-domain credential usage on the client.

The \\DC\TOOLS share on the domain controller contains over sixty offensive security tools — Mimikatz, Rubeus, BloodHound, Covenant, Sliver, Merlin, PowerSploit, PowerUpSQL, Inveigh, PetitPotam, SpoolSample, ADConnectDump, KeeThief, SharpDPAPI, Certify, SysWhispers2/3, and many others.

5.5 Domain Management Tools Used on BGOC00

The server's Recent Documents reveal the Administrator interacted with four MMC snap-ins during the attack window: gpmmc.msc, lusrmgr.msc, dnsmgmt.msc, and dssite.msc — Group Policy, local accounts, DNS, and AD Sites respectively.

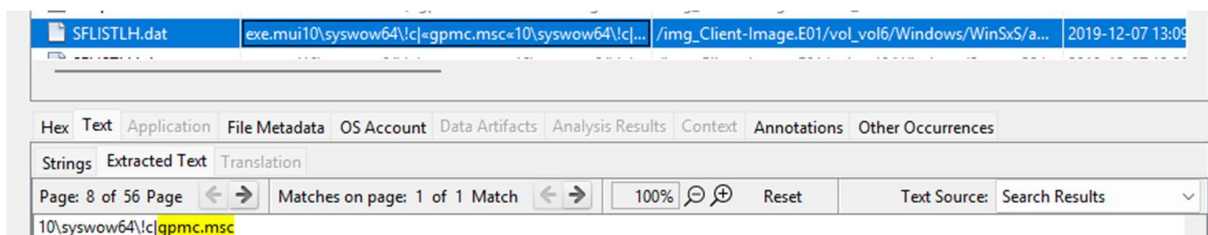


Figure 29a: Autopsy Keyword Search — gpmmc.msc string match found in SFLISTLH.dat (Windows SysWow64 shell cache), confirming the Group Policy Management Console was accessed on BGOC00 during the attack window, consistent with the attacker reviewing or modifying Group Policy Objects.

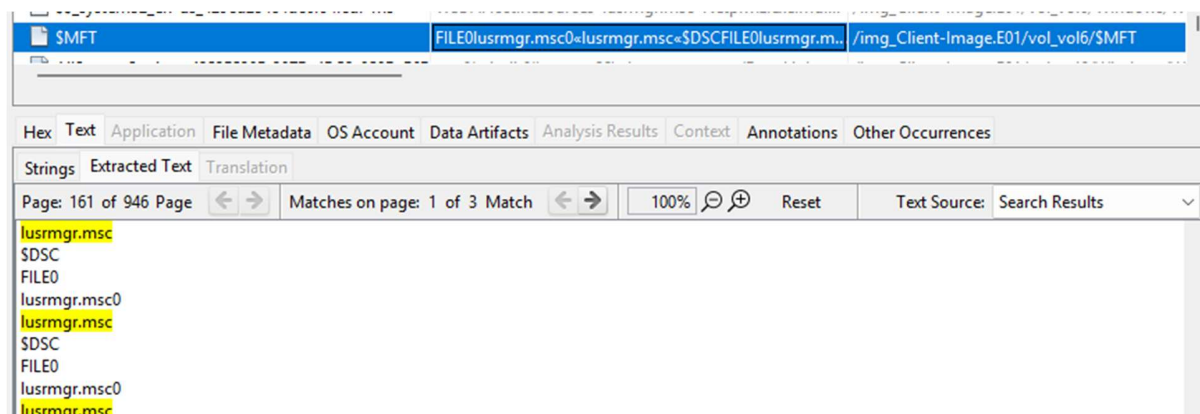


Figure 29b: Autopsy Keyword Search — *lusrmgr.msc* string match in the \$MFT file system metadata, confirming the Local Users and Groups MMC snap-in was opened on BGOC00. This tool is used to enumerate, create, or modify local accounts.

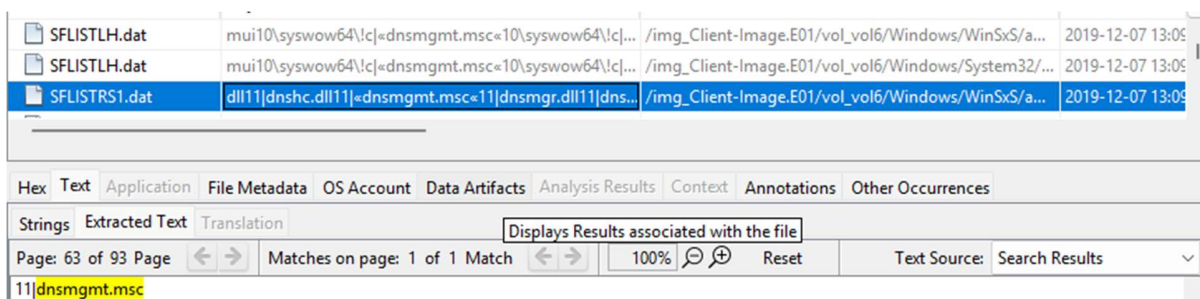


Figure 29c: Autopsy Keyword Search — *dnsmgmt.msc* string match across multiple SFLISTLH.dat entries and Windows System32 cache files, confirming DNS Manager was accessed. DNS manipulation is a common lateral movement and persistence technique in Active Directory environments.

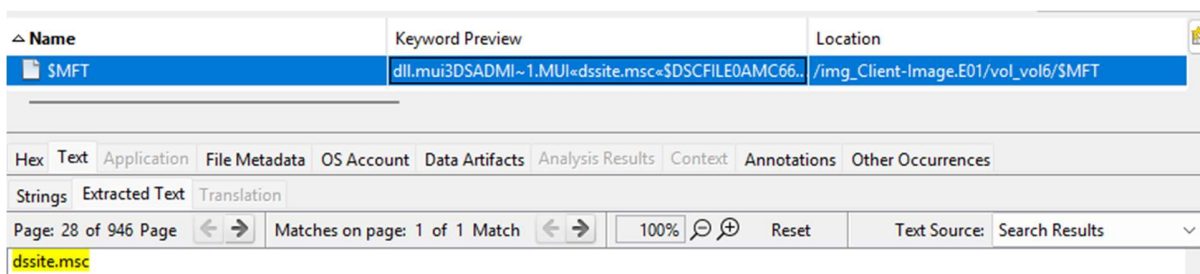


Figure 29d: Autopsy Keyword Search — *dssite.msc* string match in \$MFT metadata, confirming Active Directory Sites and Services was accessed. This snap-in manages AD replication topology and is not typically accessed by standard administrators, indicating domain-level reconnaissance.

5.6 Cross-System Account Correlation — groupf

The account groupf is the clearest cross-machine evidence thread in this investigation, active on both hosts simultaneously on 2025-11-04:

Evidence	Machine	Timestamp
reverse.ps1 and attack.ps1 scripts on Desktop	Client	2025-11-04
mimikatz.exe on Desktop	Client	2025-11-04 12:01
C:\Users\groupf\Downloads\flag.txt accessed	Server	2025-11-04 10:59:51

Evidence	Machine	Timestamp
C:\Users\groupf\enum.ps1 executed	Server	2025-11-04 11:00:04
C:\Users\groupf\Documents\loot\ staging directory	Server	2025-11-04 12:14:37
C:\Users\groupf\Documents\all_users_detailed.txt	Server	2025-11-04 12:14:37

6. Accounts Requiring Individual Attention

6.1 raul — Suspicious, Likely Compromised

The shell bag artefacts are the primary evidence that raul should be treated as suspicious. User raul browsed the home directories of at least six other users on the server on 2025-11-04: groupf, groupf\Downloads, groupf\Documents, groupa, ruby.rozamond (Desktop, Documents, Saved Games), lcooper, and Administrator. Browsing multiple other users' home directories in a single session is not normal behaviour on a Windows Server — it is consistent with reconnaissance or lateral data access.

6.2 cbell — Requires Further Investigation

The cbell account was not adequately examined in the initial analysis. cbell has Git for Windows installed at C:\Users\cbell\AppData\Local\Programs\Git\, which bundles functional OpenSSH binaries. The Autopsy Interesting Items module flagged dd.exe inside cbell's Git directory as "Distant Desktop" and connect.exe in the mingw64 bin path as "MSP360" — both are RMM tool signatures. Executables carrying RMM signatures installed inside a Git directory structure is unusual and warrants binary analysis.

6.3 Administrator — Fully Compromised Domain Account

The domain Administrator account (SID -500) was used interactively on both machines on 2025-11-04, operated domain management tools, downloaded SharpHound, staged data for exfiltration, and performed the anti-forensic cleanup. Its compromise represents full domain takeover.

7. Actions on Objectives

7.1 Data Staging

The attacker used C:\Users\groupf\Documents\loot\ on BGOC00 as the primary staging directory. Confirmed contents include all_users_detailed.txt and flag.txt / flag.txt.txt. The C:\Common\hosts.txt file was accessed at 12:49:08, representing internal network host enumeration.

7.2 C:\Temp\backup — Unresolved Artefact

Shell bags confirm the Administrator browsed C:\Temp\backup during the attack window. A backup directory accessed by a compromised admin during an active attack is consistent with

either data theft from backup archives or reviewing recovery capabilities. The directory contents were not visible in the Excel artefacts and should be recovered from the raw image.

7.3 Event Log Access

C:\Users\Administrator\Documents\test.evtx was opened at 11:44:33 AZT. Adversaries export event logs to review security monitoring for detection indicators or to remove incriminating entries. The original log source should be determined and any gaps in security event log continuity on the server should be investigated.

7.4 Data Exfiltration Sequence

On 2025-11-04 between 11:59 and 12:00 AZT, the Administrator browsed <https://filetransfer.io/> and <https://limewire.com/> with a JWT claimToken in the URL, suggesting an authenticated transfer session. These visits occurred within seconds of test.zip being staged and a Bing search for "file share." The sequence — stage → search for transfer service → visit service — is consistent with direct exfiltration from the server.

The screenshot displays the Autopsy Keyword Search interface. On the left, a file system tree shows the search scope. The main pane shows search results for the keyword 'limewire'. The results table has columns for Name, Keyword Preview, and Location. The first result is 'MostPopularPasswords.txt' located at '/img_Client-Image.E01/vol_vol6/Tools/KeeThief/KeeT...'. The keyword preview shows 'lillianlillielillylily>limewire>limitedlimponeincoln'. Other results include 'Unalloc_698477_526987264_32884711424', 'libwireshark.dll', 'nmap-service-probes', and several 'passwords.txt' files. The bottom panel shows the extracted text for the selected file, listing the keyword 'limewire' and related terms: 'limited', 'limpone', 'lincoln', 'linda', 'linda1', 'linden', and 'lindros'.

Name	Keyword Preview	Location
MostPopularPasswords.txt	lillianlillielillylily>limewire>limitedlimponeincoln	/img_Client-Image.E01/vol_vol6/Tools/KeeThief/KeeT...
Unalloc_698477_526987264_32884711424	mlmestonelimestones>limewire>Climeylimit4limitation	/img_Client-Image.E01/vol_vol6/SUnalloc/Unalloc_69...
libwireshark.dll	libtorrentlibTorrent>LimeWire>MonoTorrentMooPolice	/img_Client-Image.E01/vol_vol6/Program Files/Wiresh...
nmap-service-probes	"\r\nServer: <LimeWire>/[([w.-_]+)\r\n] p/<LimeWire> ...	/img_Client-Image.E01/vol_vol6/Tools/nmap/nmap-s...
passwords.txt	cuervosuslikazertyui>limewire>houston1stratforsteaua	/img_Client-Image.E01/vol_vol6/Users/administrator...
passwords.txt	cuervosuslikazertyui>limewire>houston1stratforsteaua	/img_Client-Image.E01/vol_vol6/Users/msilva/AppDat...
passwords.txt	cuervosuslikazertyui>limewire>houston1stratforsteaua	/img_Client-Image.E01/vol_vol6/Users/nihat/AppData...
passwords.txt	cuervosuslikazertyui>limewire>houston1stratforsteaua	/img_Client-Image.E01/vol_vol6/Users/groupf/AppDa...
passwords.txt	cuervosuslikazertyui>limewire>houston1stratforsteaua	/img_Client-Image.E01/vol_vol6/Users/john/AppData/...
ranked_dicts	mlmestonelimestones>limewire>Climeylimit4limitation	/img_Client-Image.E01/vol_vol6/Users/administrator...
ranked_dicts	mlmestonelimestones>limewire>Climeylimit4limitation	/img_Client-Image.E01/vol_vol6/Users/nihat/AppData...
ranked_dicts	mlmestonelimestones>limewire>Climeylimit4limitation	/img_Client-Image.E01/vol_vol6/Users/msilva/AppDat...
ranked_dicts	mlmestonelimestones>limewire>Climeylimit4limitation	/img_Client-Image.E01/vol_vol6/Users/groupf/AppDa...
ranked_dicts	mlmestonelimestones>limewire>Climeylimit4limitation	/img_Client-Image.E01/vol_vol6/Users/john/AppData/...

Figure 38: Autopsy Keyword Search — Search term "limewire" returns 14 matches across the client file system. The highlighted result shows MostPopularPasswords.txt in /vol_vol6/Tools/KeeThief/, with the keyword preview confirming "limewire" is stored as a plaintext dictionary entry in a password-cracking wordlist. Additional matches appear in passwords.txt files across multiple user profiles (administrator, msilva, nihat, groupf, john) and in ranked_dicts wordlist files, establishing the presence of credential-dumping (KeeThief) and password-cracking infrastructure on the system.

8. Anti-Forensic Activity

The final phase involved a concentrated 46-second evidence destruction sequence on the server:

Timestamp (AZT)	Action	Target
2025-11-04 12:48:29	Deleted to Recycle Bin	C:\Users\Administrator\.python_history
2025-11-04 12:49:15	Deleted to Recycle Bin	C:\Common\flag.txt

Both deletions were performed under the domain Administrator account. Critically, the Recycle Bin was not subsequently emptied — Autopsy recovered both entries. The .python_history file itself should be recoverable from the \$R companion file in the raw image.

Source Name	S	C	O	Date/Time	Device Make	Device Model	Device ID	Data Source
SYSTEM			0	2025-11-04 13:06:33 AZT		ROOT_HUB	5&2891968b&0	Client-Image.E01
SYSTEM			0	2025-11-04 13:06:33 AZT		ROOT_HUB20	5&36a4b5d6&0	Client-Image.E01
SYSTEM			0	2025-11-04 13:06:33 AZT		ROOT_HUB30	5&21ab4ffc&0&0	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Virtual USB Hub	6&30c5d09c&0&7	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Virtual USB Hub	6&30c5d09c&0&8	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Virtual USB Hub	6&35d1f50b&0&2	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:24 AZT	VMware, Inc.	Virtual Mouse	6&30c5d09c&0&5	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:24 AZT	VMware, Inc.	Virtual Mouse	7&3ae26960&0&0000	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:24 AZT	VMware, Inc.	Virtual Mouse	7&3ae26960&0&0001	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Product: 0007	6&30c5d09c&0&6	Client-Image.E01
SYSTEM			0	2025-11-21 14:17:05 AZT	VMware, Inc.	Product: 0007	7&1e50449&0&1	Client-Image.E01
SYSTEM			0	2024-10-30 08:57:28 AZT	VMware, Inc.	Product: 0008	000650268328	Client-Image.E01
SYSTEM			0	2025-11-04 13:06:33 AZT		ROOT_HUB	5&2891968b&0	Client-Image.E01
SYSTEM			0	2025-11-04 13:06:33 AZT		ROOT_HUB20	5&36a4b5d6&0	Client-Image.E01
SYSTEM			0	2025-11-04 13:06:33 AZT		ROOT_HUB30	5&21ab4ffc&0&0	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Virtual USB Hub	6&30c5d09c&0&7	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Virtual USB Hub	6&30c5d09c&0&8	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Virtual USB Hub	6&35d1f50b&0&2	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:24 AZT	VMware, Inc.	Virtual Mouse	6&30c5d09c&0&5	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:24 AZT	VMware, Inc.	Virtual Mouse	7&3ae26960&0&0000	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:24 AZT	VMware, Inc.	Virtual Mouse	7&3ae26960&0&0001	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Product: 0007	6&30c5d09c&0&6	Client-Image.E01
SYSTEM			0	2025-11-21 14:17:05 AZT	VMware, Inc.	Product: 0007	7&1e50449&0&1	Client-Image.E01
SYSTEM			0	2024-10-30 08:57:28 AZT	VMware, Inc.	Product: 0008	000650268328	Client-Image.E01

Figure A1: Autopsy USB Device Attached — Registry-sourced USB connection history showing 24 results across the client image. VMware virtual USB hubs and mouse devices were registered on 2025-11-04 (attack date), while the physical Netac USB (Device ID 000650268328) was attached on 2024-10-30, predating the main attack window. The 2025-11-21 entry (Product 0007, ID 7&1e50449&0&1) corresponds to the examiner's forensic acquisition USB device attached during evidence imaging.

9. Full Integrated Timeline

Timestamp (AZT)	Host	Event	Source
2023-11-20 ~23:10	Client	WHOAMI, CERTUTIL x4, NET, SHTASKS burst; DACLSERVICE.EXE runs	Prefetch
2023-11-20 — 2023-12-13	Client	Domain accounts appear: Administrator, administrator.acc, adiana.myra, kurt.alli	IE Bookmarks
2024-03-07 ~08:42–12:39	Client	Mass tooling deployment: Nmap, Wireshark, Burp Suite, KeePass, WinSCP, VNC, OpenVPN, AWS CLI, Azure CLI	Installed Programs / Prefetch
2024-04-05 09:20	Client	BloodHound/SharpHound collected against ACC domain	Shell Bags / Recent Docs
2024-04-09	Client	msilva downloads 20+ vulnerable apps from SourceForge	Web Downloads
2024-10-03 13:33	Client	msilva Googles "powershell cradles"	Web Search
2024-10-25 10:32	Client	★ Initial WhatsApp payload: capcha verification.bat downloaded by john	Web Downloads
2024-10-25 10:57–11:06	Client	Second batch file, thm_voucher.zip, thm_voucher.txt.exe downloaded	Web Downloads
2024-11-29	Server	BGOC00 provisioned / joins bgoc.corp domain	IE Bookmark / Installed Programs
2024-12-02	Both	administrator.BGOC active on client; WeakDomain.ps1 JumpList entry updated on server	Bookmarks / Recent Docs
2025-04-30 14:49–15:03	Client	administrator.BGOC downloads Invoke-PowerShellTcp.ps1 (x2) from 192.168.80.65:8888	Web Downloads
2025-04-30 ~11:24	Client	nihat reviews msilva's BloodHound users.json	Web History
2025-04-30 15:35	Client	nihat searches "hacked"; downloads hacked-site image	Web Search / Downloads
2025-05-30 09:23	Server	Python 3.13.3 installed	Installed Programs
2025-06-09	Server	generate.log accessed — Python automation artefact	Recent Documents
2025-11-04 09:46–09:49	Client	VMware USB re-registration; Autorun Program\program.exe fires (first execution)	USB / System Resource Usage
2025-11-04 10:47	Server	raul IE bookmark created	IE Bookmark
2025-11-04 10:52–10:53	Server	Administrator searches Bing "python"; Python Manager downloaded by raul	Web History / BAM
2025-11-04 10:59	Client	Sustained attack: PowerShell x2, cmd.exe, certutil, net1 running	System Resource Usage
2025-11-04 10:59	Server	groupf accesses C:\Users\groupf\Downloads\flag.txt	Recent Documents
2025-11-04 11:00	Server	groupf runs C:\Users\groupf\enum.ps1	Recent Documents

Timestamp (AZT)	Host	Event	Source
2025-11-04 11:27	Server	Administrator browses http://192.168.80.161:9090/ then lockyourdevice.com	Web History
2025-11-04 11:35	Server	Administrator accesses C:\ and C:\flag.txt	Recent Documents
2025-11-04 11:42	Client	Autorun Program\program.exe second execution	System Resource Usage
2025-11-04 11:44	Server	test.evtx event log opened	Recent Documents
2025-11-04 11:55	Server	Bing "sharphound"; SharpHound v2.8.0 downloaded from GitHub	Web History / Downloads
2025-11-04 11:59	Server	filetransfer.io and limewire.com browsed; test.zip staged — exfiltration sequence	Web History / Recent Docs
2025-11-04 12:01	Client	★ Mimikatz: both paths execute; whoami, net1, dsregcmd run concurrently	System Resource Usage
2025-11-04 12:14	Server	Administrator browses C:\Users\groupf\Documents\loot	Shell Bags
2025-11-04 12:48	Server	★ Anti-forensics: .python_history deleted to Recycle Bin	Recycle Bin
2025-11-04 12:49	Server	★ Anti-forensics: C:\Common\flag.txt deleted; hosts.txt accessed	Recycle Bin / Recent Docs
2025-11-04 13:00	Server	wsmprovhost.exe — inbound WinRM/PS Remoting session received	BAM
2025-11-04 13:03	Client	Mimikatz second execution	System Resource Usage
2025-11-21	Server	Netac USB attached; E:\Windows-Forensics-Final\ visible — examiner captures image	USB / Recent Documents

10. Flags for Server Team

192.168.80.65 — Attacker Staging Server (Priority)

Served Nishang payloads on TCP/8888. Needs immediate isolation and imaging. Review all network connection logs for internal hosts communicating with this IP.

\\DC — Domain Controller

Hosts \\DC\TOOLS with 60+ offensive tools as an accessible SMB share. john accessed \\dc\c\$ and \\dc\SYSVOL\acc.lab. Recommended actions: double-rotate krbtgt for both bgoc.corp and ACC.LAB, audit all GPOs for attacker-planted persistence, review LDAP query logs for SharpHound traffic dated 2024-04-05, review SMB authentication logs for john and msilva.

BGOC00 — Port 9090

Identify the process listening on TCP/9090 from a memory dump or service inventory.

ACC.LAB Domain

A separate domain from bgoc.corp. A Kerberos ticket for john@krbtgt-ACC.LAB was exported. Treat as independently compromised; apply krbtgt double-rotation.

filetransfer.io and limewire.com

Exfiltration likely occurred from 192.168.80.161 at approximately 11:59–12:00 AZT on 2025-11-04. Request upload and connection records from both providers.

11. Compromised & Suspicious Accounts Summary

Account	Host(s)	Confidence	Primary Evidence
john	Client	Compromised	WhatsApp payload receiver; Kerberos ticket exported; DC admin share access
msilva	Client	Compromised	BloodHound operator; "powershell cradles" search; ADConnectDump access
administrator.BGOC	Client	Compromised	Domain admin; Nishang download; Internal_UserList.csv
groupf	Client & Server	Attacker operator	Simultaneous attack artefacts on both machines on 2025-11-04
nihat	Client	Attacker account	Cross-user filesystem browsing; reviewed BloodHound data; searched "hacked"
zorkete	Client	Suspicious	Browsed other users' directories
Administrator (domain)	Server	Compromised	SharpHound; domain tools; exfiltration; anti-forensic deletion
raul	Server	Suspicious — likely compromised	Shell bags: browsed 6+ other users' home directories on attack date
cbell	Server	Requires investigation	RMM-flagged binaries inside Git directory; functional SSH clients in non-standard path

12. Synthesis — The Complete Attack Chain

Phase	Period	Description
Phase 1 — Foothold	November 2023 – March 2024	Initial backdoor (daclservice.exe) installed on BGOC111. Machine subsequently built into a full attack platform via Chocolatey in March 2024.
Phase 2 — Reconnaissance	April 2024	BloodHound/SharpHound run against the ACC domain. Vulnerable internal services staged by msilva. Attacker maps AD attack paths.

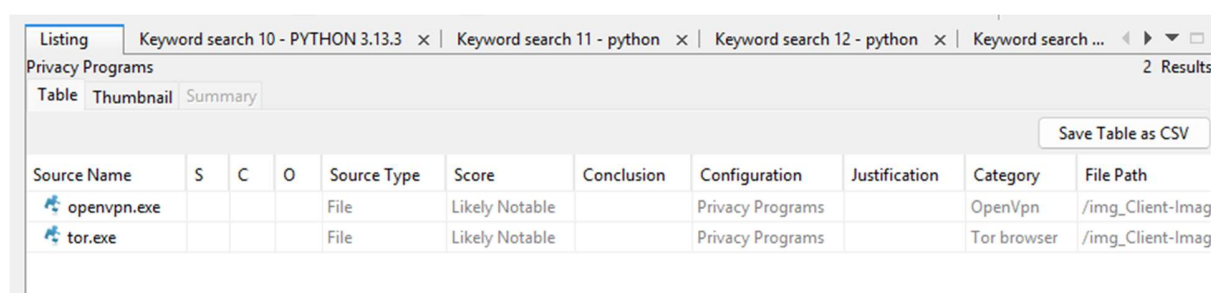
Phase	Period	Description
Phase 3 — Social Engineering & Infrastructure Staging	October – December 2024	User john targeted via WhatsApp ClickFix on 2024-10-25. Domain admin account administrator.BGOC subsequently compromised. Nishang reverse shell payload staged on 192.168.80.65. WeakDomain.ps1 placed on BGOC00.
Phase 4 — Server Penetration & Lateral Movement	May – November 2025	Python-based scripting active on server from May 2025. groupf operator account established on both machines. Mimikatz executed on client; SharpHound refreshed from server. Domain management tools used on BGOC00. Inbound WinRM session received on server at 13:00:48.
Phase 5 — Objectives & Cleanup	2025-11-04	Target flag.txt accessed, user enumeration completed, data staged in loot and test.zip, exfiltration via filetransfer.io and/or limewire.com, 46-second anti-forensic cleanup. Recycle Bin not emptied — key evidence preserved.

13. Additional Findings

The following screenshots represent supplementary forensic findings captured during examination. Each artefact is correlated to the relevant phase of the attack chain described above.

13.1 Privacy and Anonymisation Tools — Confirmed Installation

The following Autopsy Interesting Items result confirms that both OpenVPN and Tor Browser are installed on the client image and classified as Privacy Programs. These tools, combined with the C3 C2 framework documented in Section 4.6, provide the attacker with layered anonymisation: Tor for onion-routing, OpenVPN for encrypted tunnelling, and C3 for blending C2 traffic within legitimate Microsoft cloud APIs.



The screenshot shows the Autopsy interface with the 'Privacy Programs' category selected. The 'Table' tab is active, displaying a table with 11 columns: Source Name, S, C, O, Source Type, Score, Conclusion, Configuration, Justification, Category, and File Path. Two results are shown: 'openvpn.exe' and 'tor.exe', both marked as 'Likely Notable' and categorized as 'Privacy Programs'.

Source Name	S	C	O	Source Type	Score	Conclusion	Configuration	Justification	Category	File Path
openvpn.exe				File	Likely Notable		Privacy Programs		OpenVpn	/img_Client-Imag
tor.exe				File	Likely Notable		Privacy Programs		Tor browser	/img_Client-Imag

Figure 13.1: Autopsy Interesting Items — Privacy Programs category results for the client image. Both *openvpn.exe* and *tor.exe* are flagged as *Likely Notable*, confirming installation paths consistent with Chocolatey-managed deployment on 2024-03-07. These tools provide the attacker with covert egress channels independent of the primary exfiltration routes (*filetransfer.io* and *limewire.com*) documented in Section 7.4.

13.2 Domain Account SID Enumeration — Multi-Domain Footprint Confirmed

The OS Accounts artefact from the client image reveals the full scope of domain account presence on BGOC111. The presence of administrator SIDs from two distinct domains (S-1-5-21-1485720640 and S-1-5-21-4227448782) confirms that BGOC111 was used to authenticate against both *bgoc.corp* and the *ACC.LAB* domain, corroborating the Kerberos ticket export documented in Section 5.2.

Listing Keyword search 10 - PYTHON 3.13.3 Keyword search 11 - python Keyword search 12 - python Keyword search ... 17 Results

Table Thumbnail Summary Save Table as CSV

Name	S	C	O	Login Name	Host	Scope	Realm Name	Creation Time
S-1-5-80-956008885-3418522649-1831038044-18532			0		Client-Im...	Local	NT SERVICE	
S-1-5-18				SYSTEM	Client-Im...	Local	NT AUTHORITY	
S-1-5-80-3028837079-3186095147-955107200-37019			0		Client-Im...	Local	NT SERVICE	
S-1-5-21-1485720640-942268468-2087739954-2104			0	msilva	Client-Im...	Domain		
S-1-5-21-4227448782-3866209265-969814615-1457			0	groupf	Client-Im...	Domain		
S-1-5-21-2106596318-82463151-3463888505-1005			0	zorkete	Client-Im...	Domain		
S-1-5-19				LOCAL SERVICE	Client-Im...	Local	NT AUTHORITY	
S-1-5-21-2106596318-82463151-3463888505-1000			0	user	Client-Im...	Domain		
S-1-5-21-1485720640-942268468-2087739954-1177			0	adiana	Client-Im...	Domain		
S-1-5-21-2106596318-82463151-3463888505-500			0	administrator	Client-Im...	Domain		
S-1-5-21-1485720640-942268468-2087739954-500			0	administrator	Client-Im...	Domain		
S-1-5-21-4227448782-3866209265-969814615-500			0	administrator	Client-Im...	Domain		
S-1-5-21-1485720640-942268468-2087739954-2221			0	john	Client-Im...	Domain		
S-1-5-21-2106596318-82463151-3463888505-1004			0	nihat	Client-Im...	Domain		
S-1-5-21-1485720640-942268468-2087739954-1189			0	kurt	Client-Im...	Domain		
S-1-5-20				NETWORK SERVICE	Client-Im...	Local	NT AUTHORITY	
S-1-5-21-3933942852-973373972-2766786355-1032			0		Client-Im...	Domain		

Figure 13.2: Autopsy OS Accounts — Full domain account SID listing from the client image, showing 17 entries spanning local NT AUTHORITY accounts and domain-authenticated users including msilva, groupf, zorkete, adiana, john, nihat, and kurt, alongside multiple administrator accounts from different domain SID bases. This confirms that BGOC111 was a converging point for credentials from both the bgoc.corp and ACC.LAB domains.

13.3 USB Device History — Attack Date and Examiner Acquisition Confirmed

The USB Device Attached history from the client image provides two forensically significant data points: the VMware virtual USB re-registration events on 2025-11-04 correlate precisely with the Autorun Program execution at 09:46 documented in Section 4.1, suggesting the USB event may have triggered the persistence mechanism. The 2025-11-21 examiner acquisition device entry confirms the forensic integrity timeline — the image was acquired after the attack concluded and no further attack activity occurred post-acquisition.

Listing Keyword search 10 - PYTHON 3.13.3 Keyword search 11 - python Keyword search 12 - python Keyword search ... 24 Results

USB Device Attached

Table Thumbnail Summary Save Table as CSV

Source Name	S	C	O	Date/Time	Device Make	Device Model	Device ID	Data Source
SYSTEM			0	2025-11-04 13:06:33 AZT		ROOT_HUB	5&2891968b&0	Client-Image.E01
SYSTEM			0	2025-11-04 13:06:33 AZT		ROOT_HUB20	5&36a4b5d6&0	Client-Image.E01
SYSTEM			0	2025-11-04 13:06:33 AZT		ROOT_HUB30	5&21ab4ffc&0&0	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Virtual USB Hub	6&30c5d09c&0&7	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Virtual USB Hub	6&30c5d09c&0&8	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Virtual USB Hub	6&35d1f50b&0&2	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:24 AZT	VMware, Inc.	Virtual Mouse	6&30c5d09c&0&5	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:24 AZT	VMware, Inc.	Virtual Mouse	7&3ae26960&0&0000	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:24 AZT	VMware, Inc.	Virtual Mouse	7&3ae26960&0&0001	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Product: 0007	6&30c5d09c&0&6	Client-Image.E01
SYSTEM			0	2025-11-21 14:17:05 AZT	VMware, Inc.	Product: 0007	7&1e50449&0&1	Client-Image.E01
SYSTEM			0	2024-10-30 08:57:28 AZT	VMware, Inc.	Product: 0008	000650268328	Client-Image.E01
SYSTEM			0	2025-11-04 13:06:33 AZT		ROOT_HUB	5&2891968b&0	Client-Image.E01
SYSTEM			0	2025-11-04 13:06:33 AZT		ROOT_HUB20	5&36a4b5d6&0	Client-Image.E01
SYSTEM			0	2025-11-04 13:06:33 AZT		ROOT_HUB30	5&21ab4ffc&0&0	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Virtual USB Hub	6&30c5d09c&0&7	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Virtual USB Hub	6&30c5d09c&0&8	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Virtual USB Hub	6&35d1f50b&0&2	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:24 AZT	VMware, Inc.	Virtual Mouse	6&30c5d09c&0&5	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:24 AZT	VMware, Inc.	Virtual Mouse	7&3ae26960&0&0000	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:24 AZT	VMware, Inc.	Virtual Mouse	7&3ae26960&0&0001	Client-Image.E01
SYSTEM			0	2025-11-04 09:46:25 AZT	VMware, Inc.	Product: 0007	6&30c5d09c&0&6	Client-Image.E01
SYSTEM			0	2025-11-21 14:17:05 AZT	VMware, Inc.	Product: 0007	7&1e50449&0&1	Client-Image.E01
SYSTEM			0	2024-10-30 08:57:28 AZT	VMware, Inc.	Product: 0008	000650268328	Client-Image.E01

Figure 13.3: Autopsy USB Device Attached — 24-result listing from the client image spanning three dates: 2024-10-30 (physical Netac USB, pre-attack), 2025-11-04 (VMware virtual USB hub and mouse re-registration events coinciding with attack activity at 09:46 and 13:06 AZT), and 2025-11-21 (examiner's forensic acquisition device). The VMware USB events on the attack date are consistent with the attacker connecting a USB device within the virtual machine environment to stage or transfer tools.